

Functions and its Graphs

Introduction: Mathematical Rules that Shape Our World

People encounter different types of functions every day without even realizing it. When a tricycle driver charges you ₱10 for the first 3 kilometers and ₱2 for each additional kilometer, that is a **piecewise function**. When you throw a ball upward and watch it trace a curved path, that is a **quadratic function**. When you calculate how much water a cube-shaped tank can hold, that is a **cubic function**. When you find the side length of a square given its area, that is a **square root function**.

Functions are mathematical rules that connect an input to an output. Different situations require different types of functions. This reading material examines five important types of functions: **linear, quadratic, cubic, square root, cube root, and piecewise functions**. Their equations, graphs, properties, and real-life applications will be explored.

Part 1: Linear Functions

Definition and General Form

A **linear function** is a function whose graph is a straight line. Its general form is:

$$f(x)=mx+b$$

Where:

- m is the **slope** (rate of change) of the line
- b is the **y-intercept** of the line (the point where the line crosses the y-axis)

Key Properties

Property	Description
Domain	All real numbers $(-\infty, \infty)$
Range	All real numbers $(-\infty, \infty)$
Graph Shape	Straight line
Rate of Change	Constant (the slope m)
Intercepts	One x-intercept and one y-intercept

Slope and Its Meaning

The slope m tells us how steep the line is:

Slope	Direction
Positive ($m > 0$)	Line rises from left to right
Negative ($m < 0$)	Line falls from left to right
Zero ($m = 0$)	Horizontal line
Undefined	Vertical line

Example: Graphing a Linear Function

Problem: Graph $f(x)=2x+1$

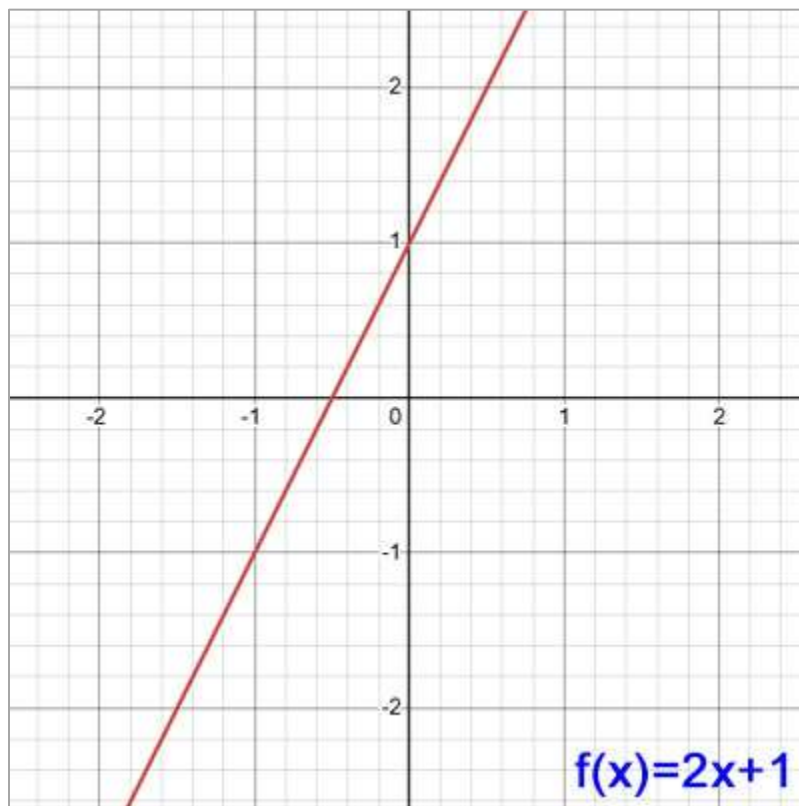
Step 1: Identify slope and y-intercept

- Slope $m=2$
- Y-intercept $b=1 \rightarrow$ point $(0,1)$

Step 2: Use slope to find another point

- From $(0,1)$, move up 2 units and right 1 unit $\rightarrow (1,3)$

Step 3: Draw the line through both points



Real-Life Applications of Linear Functions

Scenario	Linear Function	Meaning
Tricycle fare (base fare + per km)	$f(x)=10+5x$	₱10 base + ₱5 per km
Electricity bill (fixed fee + per kWh)	$B(k)=300+10k$	₱300 fixed + ₱10 per kWh
Mobile data load (₱3 per minute)	$B(t)=100-3t$	Balance decreases by ₱3/min

Part 2: Quadratic Functions

Definition and General Form

A **quadratic function** is a function whose graph is a **parabola** (a U-shaped curve). Its general form is:

$$f(x)=ax^2+bx+c$$

Where $a\neq0$. The value of a determines the direction and width of the parabola.

Key Properties

Property	Description
Domain	All real numbers $(-\infty,\infty)(-\infty,\infty)$

Property	Description
Range	Depends on vertex ($y \geq k$ or $y \leq k$)
Graph Shape	Parabola (U-shaped)
Rate of Change	Not constant; changes at every point
Vertex	Highest or lowest point
Axis of Symmetry	Vertical line through the vertex: $x = -b/2a$

Direction and Width

Value of a	Direction	Width
$a > 0$	Opens upward (vertex is minimum)	Larger a : narrower
$a < 0$	Opens downward (vertex is maximum)	Smaller a : wider

Vertex Form

Quadratic functions can also be written in **vertex form**:

$$f(x) = a(x-h)^2 + k$$

Where (h, k) is the vertex.

Example: Graphing a Quadratic Function

Problem: Graph $f(x)=x^2-4x+3$

Step 1: Find the vertex

- $a=1, b=-4$
- $x=-b/2a=2$
- $f(2)=(2)^2-4(2)+3=-1$
- Vertex: $(2,-1)(2,-1)$

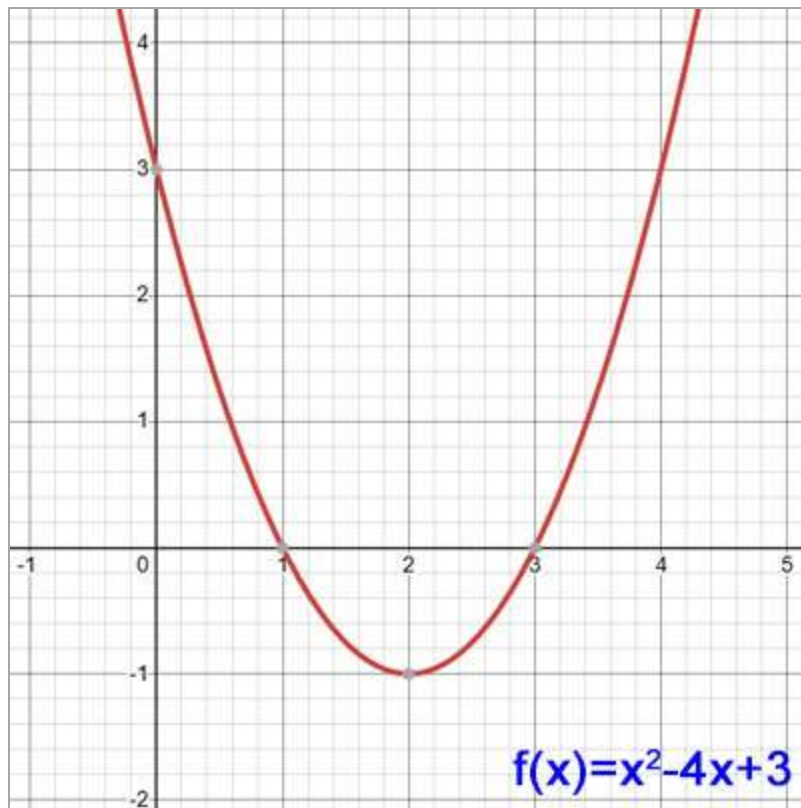
Step 2: Find y-intercept (set $x=0$)

- $f(0)=3 \rightarrow \text{point } (0,3)$

Step 3: Find x-intercepts (solve $x^2-4x+3=0$)

- $(x-1)(x-3)=0 \rightarrow x=1 \text{ and } x=3$

Step 4: Plot points and draw the parabola



Real-Life Applications of Quadratic Functions

Scenario	Quadratic Function	Meaning
Projectile motion	$h(t) = -4.9t^2 + 20t + 1.5$	Height of a thrown ball over time
Revenue optimization	$R(x) = -12x^2 + 4800$	Profit based on price changes
Maximizing area	$A(x) = -2x^2 + 30x$	Garden area with fixed fencing

Part 3: Cubic Functions

Definition and General Form

A **cubic function** is a function where the highest power of x is 3. The simplest form is:

$$f(x)=x^3$$

The general form is $f(x)=ax^3+bx^2+cx+d$ where $a \neq 0$.

Key Properties

Property	Description
Domain	All real numbers $(-\infty, \infty)(-\infty, \infty)$
Range	All real numbers $(-\infty, \infty)(-\infty, \infty)$
Graph Shape	S-shaped curve
Intercepts	Passes through the origin $(0,0)(0,0)$
Symmetry	Odd function: $f(-x)=-f(x)$ (symmetric about the origin)
End Behavior	As $x \rightarrow \infty, y \rightarrow \infty$; as $x \rightarrow -\infty, y \rightarrow -\infty$

Table of Values for $y=x^3$

x	-2	-1	0	1	2
y	-8	-1	0	1	8

Transformations of Cubic Functions

Transformation	Example	Effect
Vertical shift	$f(x)=x^3+2$	Moves graph up 2 units
Horizontal shift	$f(x)=(x-3)^3$	Moves graph right 3 units
Reflection across x-axis	$f(x)=-x^3$	Flips graph upside down
Vertical stretch	$f(x)=2x^3$	Makes graph steeper
Vertical compression	$f(x)=1/2 x^3$	Makes graph flatter

Real-Life Applications of Cubic Functions

Scenario	Cubic Function	Meaning
Volume of a cube	$V(s)=s^3$	Volume as a function of side length

Packaging design

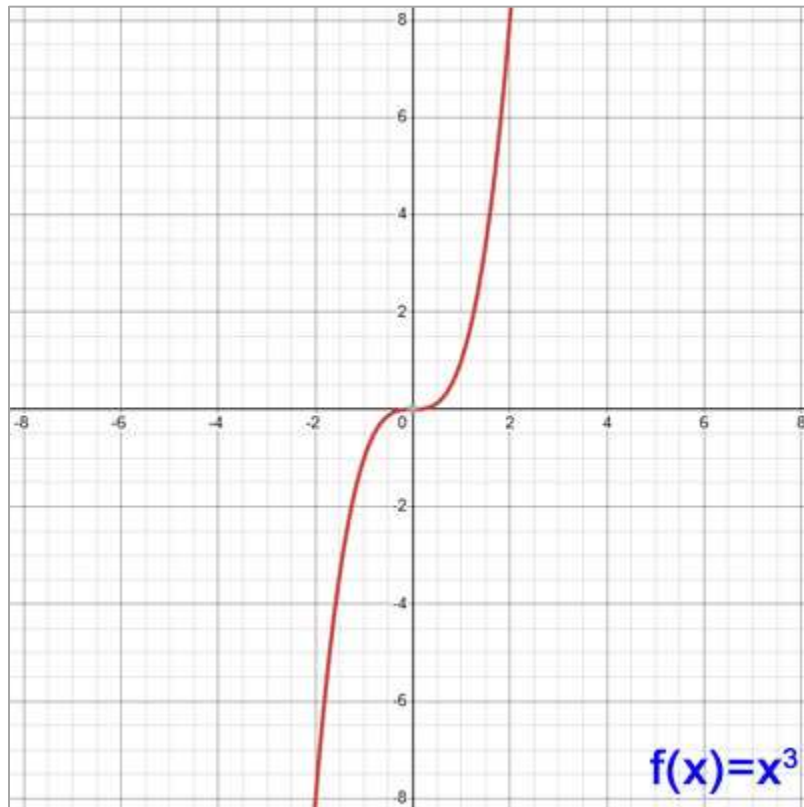
$$V(x)=x(10-2x)(8-2x)$$

Volume of a box from
folded cardboard

Population growth (some
models)

$$P(t)=at^3+bt^2+ct+d$$

Certain growth patterns



Part 4: Square Root Functions

Definition and General Form

A **square root function** involves the square root of the input variable. The simplest form is:

$$f(x) = \sqrt{x}$$

This can also be written in exponential form as $f(x) = x^{1/2}$.

Key Properties

Property	Description
Domain	$x \geq 0$ (non-negative real numbers)
Range	$y \geq 0$ (non-negative real numbers)
Graph Shape	Curves upward from the origin
Intercepts	Passes through the origin $(0,0)$
Symmetry	Neither even nor odd
End Behavior	As $x \rightarrow \infty$, $y \rightarrow \infty$ (increases slowly)

Table of Values for $y = x^{1/2}$

x 0 1 4 9 16

y 0 1 2 3 4

Transformations of Square Root Functions

General form: $f(x)=a(b(x-h))^{1/2}+k$

Transformation	Example	Effect
Vertical shift	$f(x)=\sqrt{x+3}$	Moves graph up 3 units
Horizontal shift	$f(x)=\sqrt{(x-2)}$	Moves graph right 2 units
Reflection across x-axis	$f(x)= -\sqrt{x}$	Flips graph upside down
Vertical stretch	$f(x)=2\sqrt{x}$	Makes graph taller
Reflection across y-axis	$f(x)=\sqrt{-x}$	Domain changes to $x\leq 0$

Real-Life Applications of Square Root Functions

Scenario	Square Root Function	Meaning
Side length from area	$s=\sqrt{A}$	Find side of square given area
Pendulum period	$T=2\pi\sqrt{(L/g)}$	Time period of pendulum depends

Scenario

Square Root Function

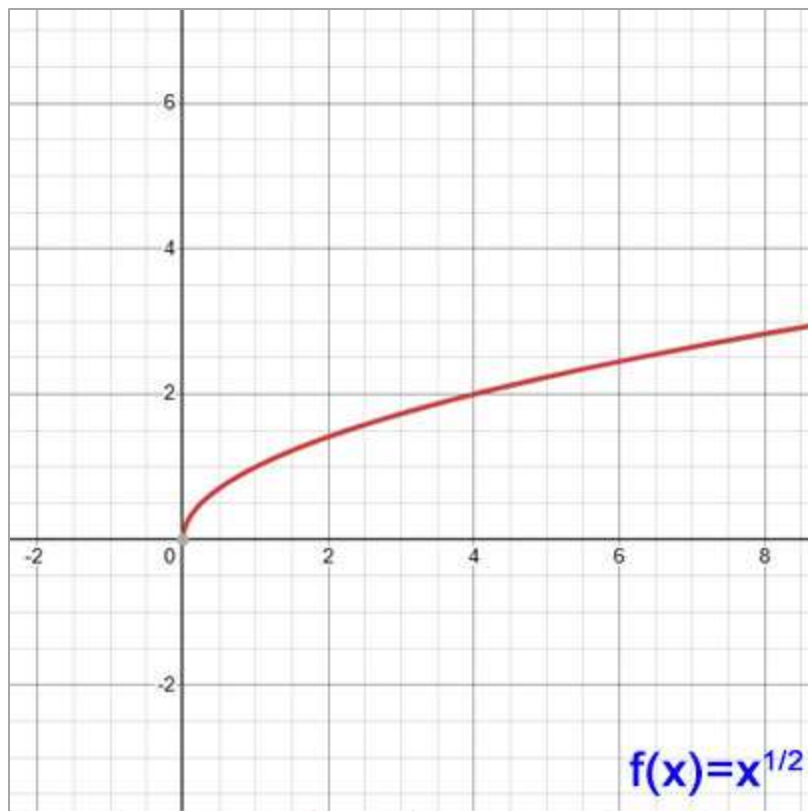
Meaning

on square root of length

Speed from kinetic energy

$$v = \sqrt{2KE / m}$$

Speed related to square root
of energy



Part 5: Cube Root Functions

Definition and General Form

A **cube root function** involves the cube root of the input variable. The simplest form is:

$$f(x) = \sqrt[3]{x}$$

This can also be written in exponential form as $f(x) = x^{1/3}$.

Key Properties

Property	Description
Domain	All real numbers $(-\infty, \infty)$
Range	All real numbers $(-\infty, \infty)$
Graph Shape	S-shaped curve (similar to cubic but rotated)
Intercepts	Passes through the origin $(0,0)$
Symmetry	Odd function: $f(-x)=-f(x)$ (symmetric about the origin)
End Behavior	As $x \rightarrow \infty, y \rightarrow \infty$; as $x \rightarrow -\infty, y \rightarrow -\infty$

Table of Values for $y=x^{1/3}$

x -8 -1 0 1 8

y -2 -1 0 1 2

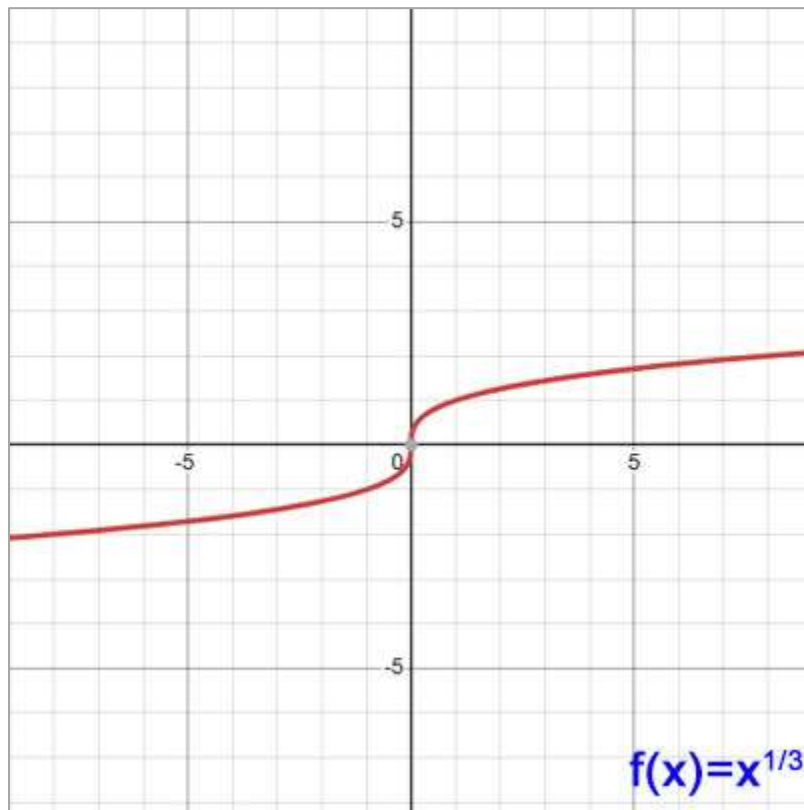
Transformations of Cube Root Functions

General form: $f(x)=a(b(x-h))^{1/3}+k$

Transformation	Example	Effect
Vertical shift	$f(x)=\sqrt[3]{x+2}$	Moves graph up 2 units
Horizontal shift	$f(x)=\sqrt[3]{(x-3)}$	Moves graph right 3 units
Reflection across x-axis	$f(x)=-\sqrt[3]{x}$	Flips graph upside down
Vertical stretch	$f(x)=3\sqrt[3]{x}$	Makes graph steeper
Reflection across y-axis	$f(x)=\sqrt[3]{(-x)}$	Reflects horizontally

Real-Life Applications of Cube Root Functions

Scenario	Cube Root Function	Meaning
Side length from volume	$s = \sqrt[3]{V}$	Find cube side given volume
Radius from sphere volume	$r = \sqrt[3]{(3V/4\pi)}$	Find radius given volume
Dilution calculations	$c = \sqrt[3]{(m/V)}$	Concentration in some models



Part 6: Piecewise Functions

Definition

A **piecewise function** is a function that uses different formulas (or "pieces") for different intervals of the input variable. Instead of one rule for all inputs, it has multiple rules depending on the value of x . The absolute value function is a simple example of a piecewise function.

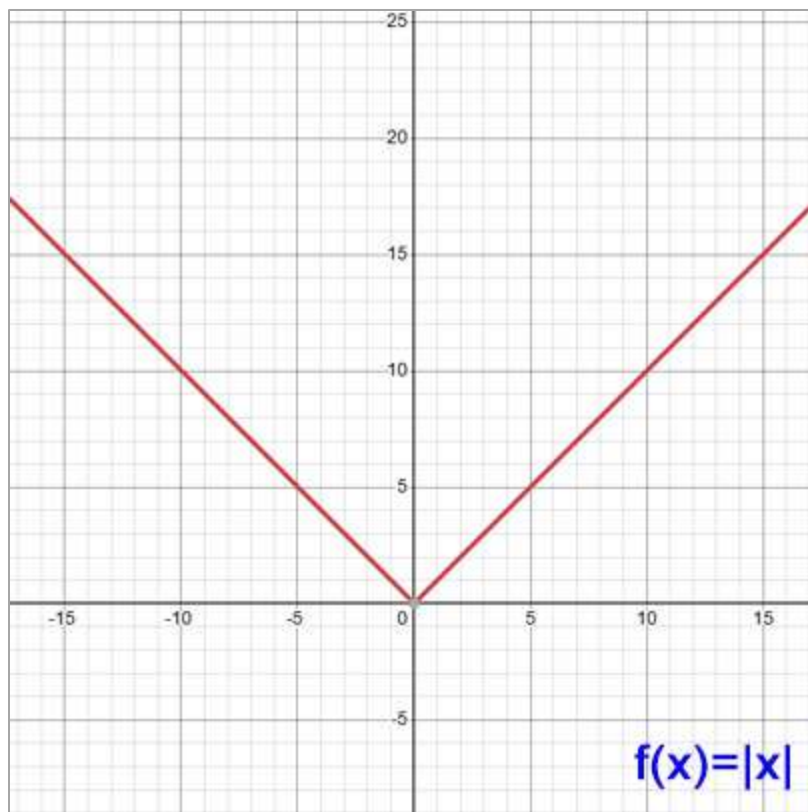
The Absolute Value Function

The absolute value of a number is its distance from zero on the number line. It is always non-negative.

$$|x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases}$$

The graph is V-shaped with the vertex at the origin $(0,0)$. The left side is a line with slope -1 , and the right side is a line with slope 1 .

Key Insight: Any absolute value function can be rewritten as a piecewise function with two pieces — one for when the expression inside is negative, and one for when it is non-negative.



Evaluating Piecewise Functions

To evaluate a piecewise function, determine which condition the input satisfies, then use the corresponding formula.

Example 1: Taxi Fare – ₱35 for first 1.5 km, ₱8 per km for next 10.5 km, ₱20 per km beyond 12 km.

$f(d) =$

35, when $0 < d \leq 1.5$

$35 + 8(d - 1.5)$, when $1.5 < d \leq 12$

$119 + 20(d - 12)$, when $d > 12$

- $d = 1.2 \text{ km} \rightarrow f(1.2) = 35 \text{ pesos}$
- $d = 5 \text{ km} \rightarrow f(5) = 35 + 8(3.5) = 63 \text{ pesos}$
- $d = 15 \text{ km} \rightarrow f(15) = 119 + 20(3) = 179 \text{ pesos}$

Example 2: Overtime Pay – ₱80/hour for up to 6 hours, ₱100/hour beyond 6 hours.

$$f(h) =$$

$$80h, \text{ when } 0 \leq h \leq 6$$

$$480 + 100(h - 6), \text{ when } h > 6$$

- $h = 4 \text{ hours} \rightarrow f(4) = 320 \text{ pesos}$
- $h = 8 \text{ hours} \rightarrow f(8) = 480 + 200 = 680 \text{ pesos}$

Graphing Piecewise Functions

To graph a piecewise function:

1. Graph each piece on its specified interval
2. Use an **open circle** (○) to indicate an endpoint is **not included**
3. Use a **closed circle** (●) to indicate an endpoint **is included**
4. Check for continuity or jumps at the breakpoints

Example: Graph of $f(x)$

$$f(x) =$$

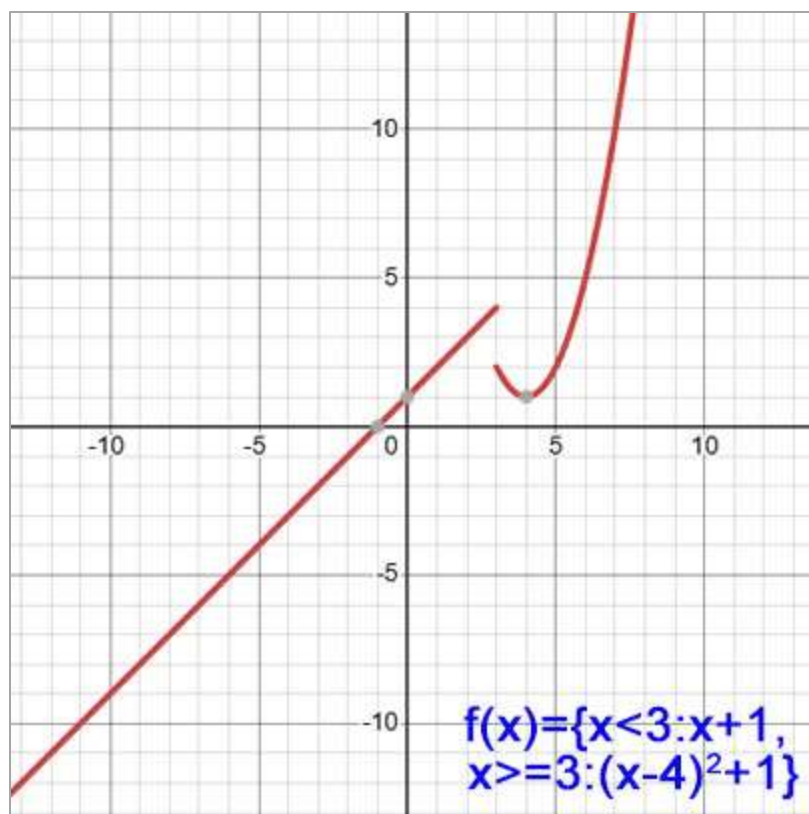
$$x + 1, \text{ when } x < 3$$

$$(x - 4)^2 + 1, \text{ when } x \geq 3$$

At $x = 3$:

- Left side: $f(3^-) = 3 + 1 = 4$
- Right side: $f(3) = (3 - 4)^2 + 1 = 2$

Since $4 \neq 2$, the graph has a jump discontinuity at $x = 3$.



Real-Life Applications of Piecewise Functions

Scenario	Breakpoints	Rule
Taxi Fare	1.5 km, 12 km	Flat ₱35, then ₱8/km, then ₱20/km
Parking Fee	1 hour, 5 hours	₱100 first hour, ₱75/additional hour up to 5 hrs, flat ₱500 beyond
Water Bill	10 m ³ , 30 m ³	₱20/m ³ , then ₱30/m ³ , then ₱50/m ³
Internet Data	3 GB, 7 GB	Free, then ₱50/GB, then ₱70/GB

Scenario	Breakpoints	Rule
Bulk Discount	40 pieces	₱15/piece, flat ₱550 for 40, then ₱550 + ₱10/excess
Community Tax	₱100,000, ₱200,000	₱0, then 2% over ₱100k, then 7% over ₱200k

Writing Piecewise Functions from Word Problems

Step 1: Identify the different intervals based on the input variable.

Step 2: Determine the formula or rule for each interval.

Step 3: Write the function using piecewise notation.

Step 4: Test with sample values to verify correctness.

Example: Parking Fee – ₱100 for first hour, ₱75 for each additional hour up to 5 hours, flat ₱500 for more than 5 hours.

$f(t) = 100$ for $0 < t \leq 1$, $f(t) = 100 + 75(t - 1)$ for $1 < t \leq 5$, and $f(t) = 500$ for $t > 5$

or

$f(t) = 100, 0 < t \leq 1; 100 + 75(t - 1), 1 < t \leq 5; 500, t > 5$

Key Terms

Term	Definition
Linear Function	A function whose graph is a straight line; general form $f(x)=mx+b$, where m is the slope and b is the y-intercept
Slope	The rate of change of a linear function; represents steepness and direction (positive = rising, negative = falling, zero = horizontal, undefined = vertical)
y-intercept	The point where the graph crosses the y-axis; found by evaluating $f(0)$
Quadratic Function	A function whose graph is a parabola; general form $f(x)=ax^2+bx+c$, where $a \neq 0$
Parabola	The U-shaped graph of a quadratic function; opens upward if $a>0$, opens downward if $a<0$
Vertex	The highest or lowest point of a parabola; the maximum or minimum of a quadratic function
Axis of Symmetry	A vertical line that passes through the vertex of a parabola
Cubic Function	A function where the highest power of x is 3; simplest form $f(x)=x^3$; graph is an S-shaped curve
Inflection Point	The point where a cubic graph changes concavity
Square Root Function	A function defined as $f(x)=\sqrt{x}$; domain is $x \geq 0$
Cube Root	A function defined as $f(x)=\sqrt[3]{x}$; domain and range are all real

Term	Definition
Function	numbers; graph is an S-shaped curve passing through the origin
Absolute Value Function	A function that gives the distance of a number from zero on the number line, resulting in a non-negative value regardless of the input
Piecewise Function	A function that uses different formulas for different intervals of the input
Breakpoint	The value of x where the rule changes in a piecewise function
Continuity	The absence of a jump, gap, or break in the graph of a function
Jump Discontinuity	A break in the graph where the left-hand limit and right-hand limit exist but are not equal; common in piecewise functions
Domain	The set of all possible input values (x) for which a function is defined
Range	The set of all possible output values (y) that a function can produce
End Behavior	The behavior of a graph as x approaches positive or negative infinity (∞ or $-\infty$)
Transformation	Changes made to a parent function that shift, stretch, compress, or reflect its graph (e.g., vertical shift, horizontal shift, reflection)

Review Questions

Write your answers on a separate sheet of paper.

Linear Functions

1. A tricycle driver charges a base fare of ₱10 plus ₱5 per kilometer traveled. Write a linear function $F(x)$ for the total fare when traveling x kilometers. How much would it cost to travel 8 kilometers?
2. A household pays a fixed monthly connection fee of ₱300 and ₱10 per kilowatt-hour of electricity consumed. Write a linear function $B(k)$ for the monthly bill based on k kilowatt-hours. Calculate the bill for 150 kWh.

Quadratic Functions

3. A ball is thrown upward and its height is modeled by $h(t) = -4.9t^2 + 20t + 1.5$, where t is time in seconds and $h(t)$ is height in meters. At what time does the ball reach its maximum height? What is the maximum height?
4. A small halo-halo stand sells each cup for ₱40 and serves 120 customers per day. For every ₱2 increase in price, they lose 6 customers. Write a quadratic function $P(x)$ for the daily revenue where x is the number of ₱2 price increases. Find the maximum revenue.

Cubic Functions

5. A cube-shaped water tank has a volume of 216 cubic meters. What is the length of one side? What is the surface area of the tank?
6. A carpenter is making small wooden cubes. If each cube has a side length of 4 cm, what is its volume?

Square Root Functions

7. A farmer has a square plot of land with an area of 225 square meters. What is the length of one side?

8. A learner is solving a math problem: Find the value of y if $x=49$ using the square root function $y=\sqrt{x}$.

Cube Root Functions

9. A toy maker has a cube-shaped box with a volume of 1,000 cubic centimeters. What is the length of each side?
10. A cold storage unit is designed in a cube shape. If the side length is 5 meters, what is its volume? If the volume is 64 m^3 , what is the side length?

Piecewise Functions

11. A tricycle ride costs ₱10.00 for the first 3 kilometers, and each additional full kilometer adds ₱2.00 to the fare. Let d be the distance in kilometers. Write the piecewise function for the total fare $f(d)$.
12. A rental car charges ₱100 for the first three hours and ₱20 for each additional hour (or fraction thereof) after that. If rented for more than 10 hours, a flat fee of ₱500 is charged. Write the piecewise function for the rental fee $R(t)$ where t is the number of hours.
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Answer Key

Linear Functions

1. $F(x)=10+5x$; $F(8)=10+5(8)=\text{₱}50$
2. $B(k)=300+10k$; $B(150)=300+10(150)= \text{₱}1,800$

Quadratic Functions

3. $t = -b/2a = -20/((2)(-4.9)) \approx 2.04$ seconds;
 $h(2.04) = -4.9(2.04)^2 + 20(2.04) + 1.5 \approx 21.9$ meters
4. $P(x) = (40 + 2x)(120 - 6x) = -12x^2 + 4800$;
Maximum profit = ₦4,800 at $x=0$ (no price increase)

Cubic Functions

5. Side length $s = \sqrt[3]{216} = 6$ meters ; Surface area $= 6s^2 = 6 \times 36 = 216 \text{ m}^2$
6. Volume $V = s^3 = 4^3 = 64 \text{ cm}^3$

Square Root Functions

7. Side length $s = \sqrt{225} = 15$ meters
8. $y = \sqrt{49} = 7$

Cube Root Functions

9. Side length $s = \sqrt[3]{1000} = 10 \text{ cm}$
10. Volume $V = 5^3 = 125 \text{ m}^3$; Side length $s = \sqrt[3]{64} = 4 \text{ m}$

Piecewise Functions

11. $f(d) = \{10, 0 < d \leq 3; 10 + 2(d - 3), d > 3\}$
12. $R(t) = \{100, 0 < t \leq 3; 100 + 20(t - 3), 3 < t \leq 10; 500, t > 10\}$